

Introduction to Microeconomics
Lecture 9—Consumer Theory: Introduction

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The Law of Demand

People do less of what they want to do as the cost of doing it rises.

 The cost of an activity, good, or service involves not just monetary costs, but nonmonetary costs as well.

- The goal is to provide a theory of consumer choice based on some reasonable assumptions about the consumer.
- What we will study is part of the workhorse models in Microeconomics and also one of the foundations of modern Macroeconomic Theory.
- Our approach will be to study the consumer embedded in a market Economy, where there are prices for goods and services, as well as wealth constraints.
- However, at its core, the approach is quite general and can be used to analyze a wide variety of problems, not only limited to what is traditionally understood as Economics, but from other social sciences as well.

The Rational Utility Maximizer

- As an introduction and for pedagogic purposes, we will start with a very intuitive (although somewhat incorrect) model of the consumer.

Pepsi consumption (cans/week)	Utility from Pepsi consumption (utils/week)
0	0
1	10
2	15
3	19
4	22
5	24
6	25

- Utility (a measure of happiness/comfort/satisfaction) increases with consumption.
- Beyond some point, the additional utility from extra consumption declines.

Marginal Utility

The marginal utility from consuming a good is the additional utility that results from consuming an additional unit of the good.

Law of diminishing marginal utility

The law of diminishing marginal utility says that as consumption of a good increases beyond some point, the additional utility that results from an additional unit of the good declines.

Possible exceptions: unfamiliar/new food or experiences; addictive goods.

Pepsi cans	Utility	Marginal utility
0	0	
1	10	10
2	15	5
3	19	4
4	22	3
5	24	2
6	25	1

Optimal combination of goods

The **optimal combination of goods** is that combination that yields the highest total utility among all the affordable combinations.

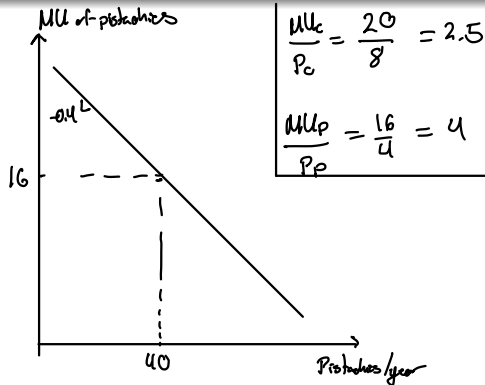
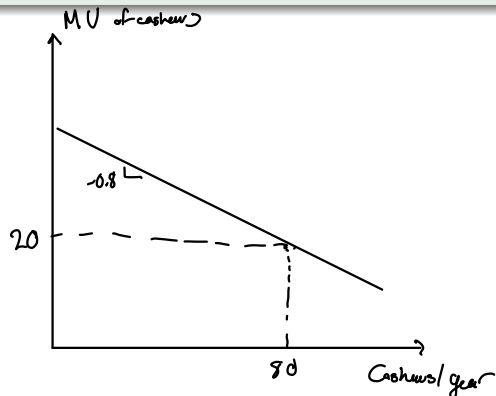
When the quantity of each good can be varied continuously, we have:

The Rational Spending rule

Spending should be allocated across goods so that the marginal utility per unit of money is the same for each good.

Is Nazerke maximizing her utility from consuming cashews or pistachios?

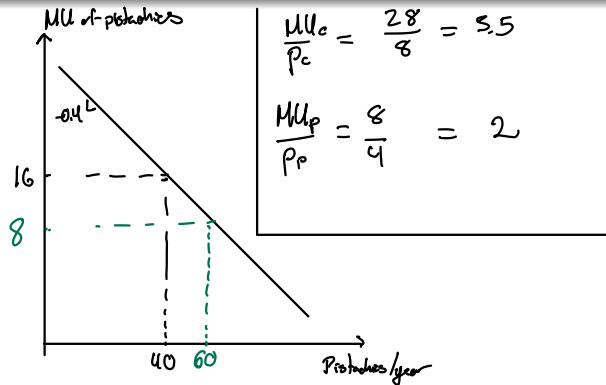
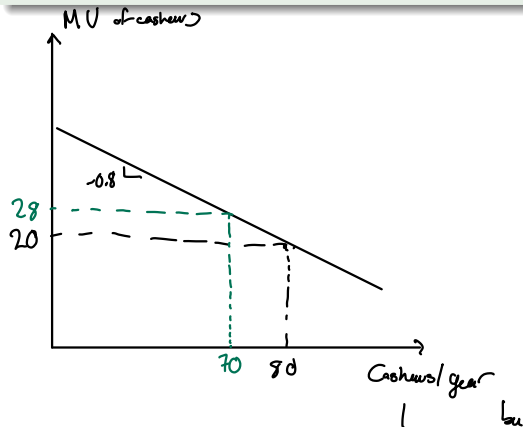
Cashew nuts sell for \$8 per kg and pistachios sell for \$4 per kg. Nazerke has a budget of \$800 per year to spend on nuts, and her marginal utility from consuming each type of nut varies with the amount consumed as shown in below. If she is currently buying 80 kilograms of cashews and 40 kilograms of pistachios each year, is she maximizing her utility?



Is she spending all her budget: $\$ (80) + 4(40) = 800$

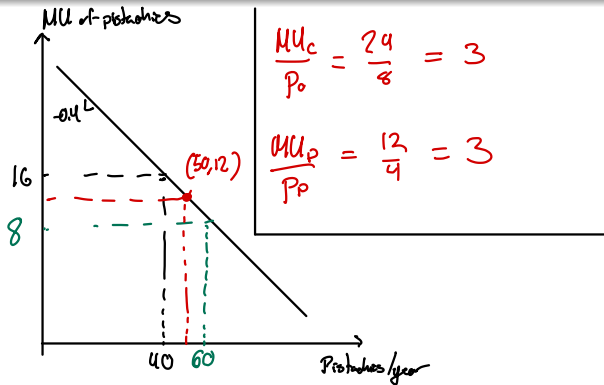
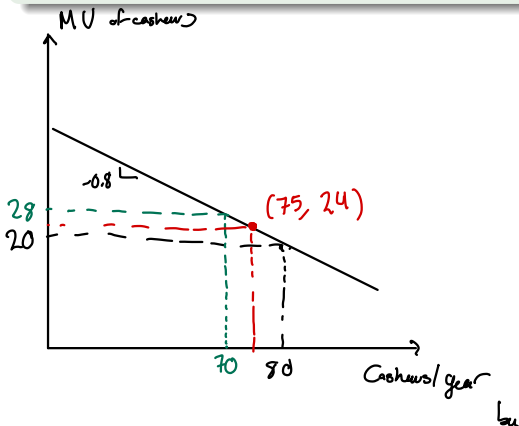
Is Nazerke maximizing her utility from consuming cashews or pistachios? Part 2

Cashew nuts sell for \$8 per kg and pistachios sell for \$4 per kg. Nazerke has a budget of \$800 per year to spend on nuts, and her marginal utility from consuming each type of nut varies with the amount consumed as shown in below. If she is currently buying 70 kilograms of cashews and 60 kilograms of pistachios each year, is she maximizing her utility?



Is Nazerke maximizing her utility from consuming cashews or pistachios? Part 3

Cashew nuts sell for \$8 per kg and pistachios sell for \$4 per kg. Nazerke has a budget of \$800 per year to spend on nuts, and her marginal utility from consuming each type of nut varies with the amount consumed as shown in below. If she is currently buying 75 kilograms of cashews and ~~80~~ kilograms of pistachios each year, is she maximizing her utility?



The rational spending rule for two goods

Let there be two goods, call them x and y . Let MU_x and MU_y denote the marginal utility of each good. Let p_x and p_y denote their prices. Then the consumer is maximizing her utility when

$$\frac{MU_x}{p_x} = \frac{MU_y}{p_y}.$$

(As long as all your budget is spent)

Buyer's Problem: \$300 available

sweaters s $p_s = \$25$				jeans j $p_j = \$50$			
$Q(s)$	$U(s)$	$MU(s)$	MU/p_s	$Q(j)$	$U(j)$	$MU(j)$	MU/p_j
0	0			0			
<u>1</u>	100	100	<u>4</u>	<u>1</u>	160	160	<u>3.2</u>
<u>2</u>	185	85	<u>3.4</u>	2	310	150	3
<u>3</u>	260	75	<u>3</u>	<u>3</u>	410	100	<u>2</u>
4	325	65	2.6	4	490	80	1.6
5	385	60	2.4	5	520	30	0.6
<u>6</u>	425	50 50	2.0 2.0	6	530	10	0.2
7	480	55 45	2.2 1.8	7	533	3	0.06
8	520	40	1.6	8	535	2	0.04

6 sweaters

3 jeans

$$6 \times 25 + 3 \times 50 = 300$$

$$\frac{MU_s}{p_s} = 2$$

$$\frac{MU_j}{p_j} = 2$$

⚠ Remark (3,2) satisfies $\frac{MU_s}{p_s} = \frac{MU_j}{p_j}$ but you are not exhausting your budget

- First \$25 → 1 sweater, \$25 remains
- next \$25 → 2 sweaters, \$25 remains
- next \$50 → 1 jeans, \$200 remains
- next \$25 → 3 sweaters, \$175 remains

next \$50 → 2 jeans → \$125

The rational spending rule for n goods

Let i index the goods, and let MU_i denote their marginal utilities. Let p_i denote their prices, $i = 1, \dots, n$. Then the consumer is maximizing her utility when

$$\frac{MU_i}{p_i} = \frac{MU_j}{p_j}, \text{ for all } i, j.$$

(Until all your budget is spent)

$$\frac{MU_1}{p_1} = \frac{MU_2}{p_2} = \dots = \frac{MU_n}{p_n}$$